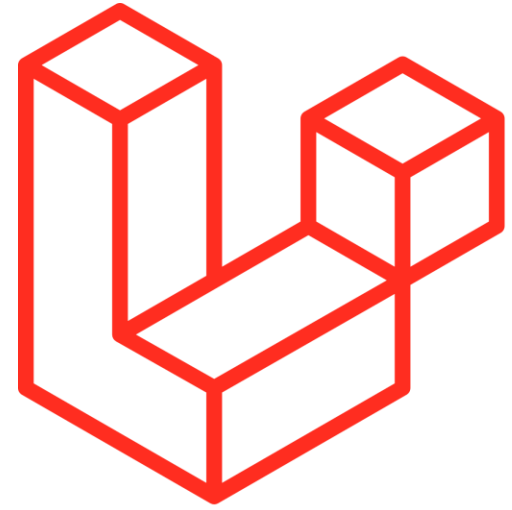


Laravel CRUD (Web)



Oum Saokosal

Master's Degree in Information Systems, South Korea

Tel/Telegram: 012 252 752

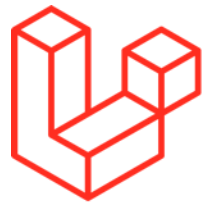
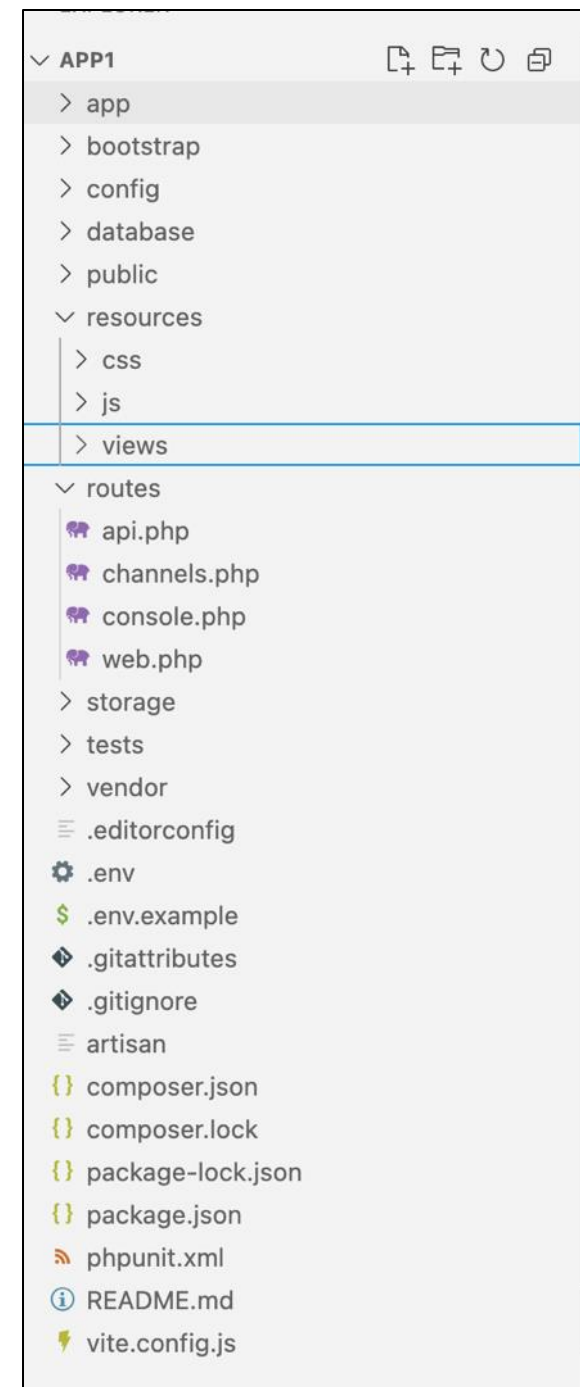
Create a Laravel project

- Make sure you have [PHP](#) and [Composer](#) installed:

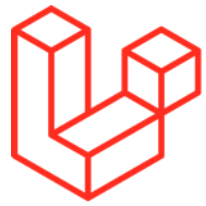
```
mac — -bash — 71x9
[(base) Macs-iMac:~ mac$ php --version
PHP 8.3.7 (cli) (built: May 7 2024 16:35:26) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.3.7, Copyright (c) Zend Technologies
    with Zend OPcache v8.3.7, Copyright (c), by Zend Technologies
[(base) Macs-iMac:~ mac$ composer --version
Composer version 2.6.5 2023-10-06 10:11:52
(base) Macs-iMac:~ mac$
```

- Create a laravel project:

- composer create-project [laravel/laravel](#) **app1**
- **cd** **app1**
- **code .**
- **php artisan serve**



PostgreSQL



- Make sure you download [PostgreSQL](#)
- During the setup, password: 123456789 and port: 5432

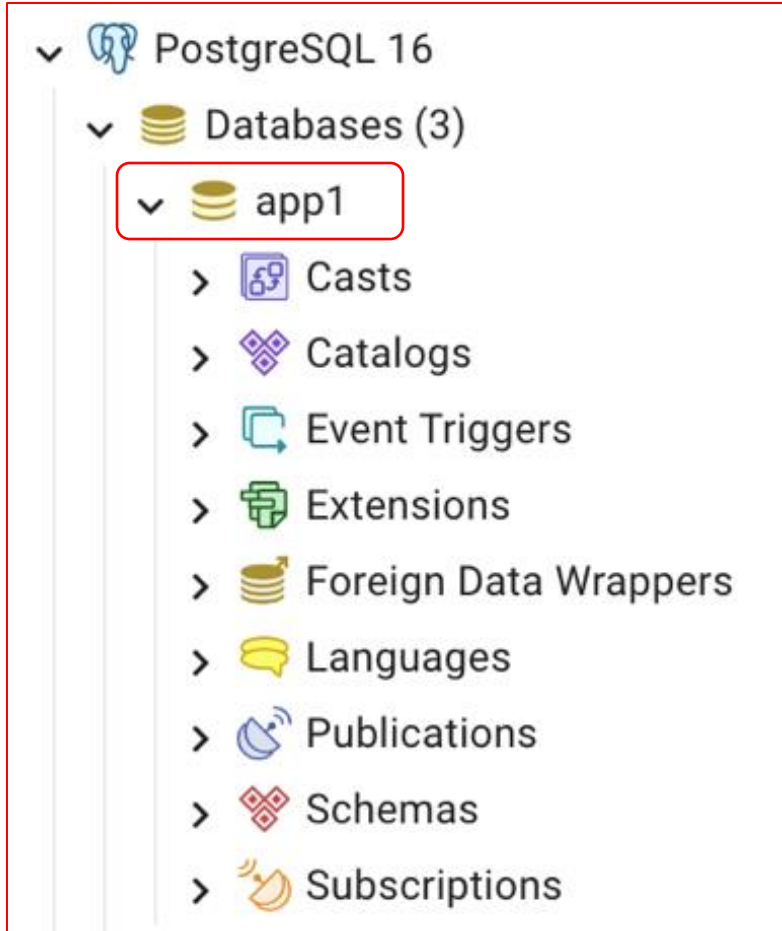
The screenshot shows the 'Password' step of the PostgreSQL setup. The window title is 'Setup'. It has a green arrow icon in the top right corner. The text 'Please provide a password for the database superuser (postgres).' is displayed. Below it, there are two input fields: 'Password' and 'Retype password', both containing asterisks. At the bottom, there are three buttons: '< Back', 'Next >', and 'Cancel'. The 'InstallBuilder' logo is in the bottom left corner.

The screenshot shows the 'PostgreSQL Port' step of the PostgreSQL setup. The window title is 'Setup'. It has a plus icon in the top right corner. The text 'The installer will initialize the PostgreSQL database server with this port (leave as-is if you are unsure what to enter)' is displayed. Below it, there is an input field for 'PostgreSQL Server Port' with the value '5432' entered. At the bottom, there are three buttons: '< Back', 'Next >', and 'Cancel'. The 'InstallBuilder' logo is in the bottom left corner.

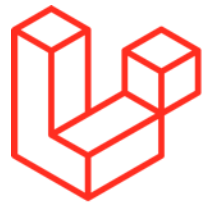
Create a database in Postgres



- Create a database and name it: **app1**



Config .env



- Config Postgres, port: 5432 and Password: 123456789

```
DB_CONNECTION=pgresql
DB_HOST=127.0.0.1
DB_PORT=5432
DB_DATABASE=app1
DB_USERNAME=postgres
DB_PASSWORD=123456789
```

Migration



1

```
php artisan make:migration create_students_table
```

2

```
// database/migrations/xxxx_xx_xx_create_students_table.php
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

class CreateStudentsTable extends Migration{
    public function up(): void{
        Schema::create('students', function (Blueprint $table) {
            $table->id('sid'); // primary key
            $table->string('name');
            $table->timestamps();
        });
    }

    public function down(): void {
        Schema::dropIfExists('students');
    }
}
```



3

```
php artisan migrate
```

▼	📄	students
▼	📄	Columns (4)
	📄	sid
	📄	name
	📄	created_at
	📄	updated_at
>	▶◀	Constraints
>	🌐	Indexes
>	🔒	RLS Policies
>	🏠	Rules
>	⚡	Triggers

Model



1

```
php artisan make:model Student
```

2

```
// app/Models/Student.php
```

```
namespace App\Models;
```

```
use Illuminate\Database\Eloquent\Model;
```

```
class Student extends Model{
```

```
    //protected $table = 'tbl_student'; // custom table name
```

```
    protected $primaryKey = 'sid'; // custom primary key
```

```
    // protected $incrementing = false; //set to false if the primary key is not incrementing
```

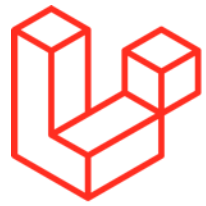
```
    protected $fillable = ['name'];
```

```
    // public $timestamps = false; //set to false if you don't want created_at & updated_at
```

```
}
```

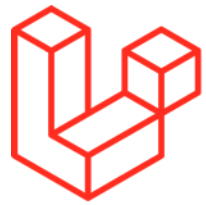


Web Controller



```
php artisan make:controller StudentController --resource
```

```
✓ APP1
  ✓ app
    > Console
    > Exceptions
    ✓ Http
      ✓ Controllers
        🐘 Controller.php
        🐘 StudentController.php
      > Middleware
      🐘 Kernel.php
    > Models
```

```
// app/Http/Controllers/StudentController.php
```

```
namespace App\Http\Controllers;
```

```
use App\Models\Student;
```

```
use Illuminate\Http\Request;
```

```
class StudentController extends Controller{
```

```
    public function index(){
```

```
        //$students = Student::all();
```

```
        $students = Student::orderBy('sid')->get();
```

```
        return view('students.index', ['students' => $students]);
```

```
    }
```

```
    public function create(){
```

```
        return view('students.create');
```

```
    }
```

```
    public function store(Request $request){
```

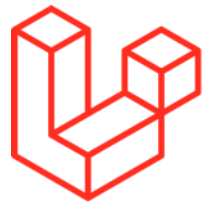
```
        $request->validate(['name' => 'required']);
```

```
        Student::create($request->all());
```

```
        return redirect()->route('students.index')->with('success', 'Student added.');
```

```
    }
```





```
public function show($id){  
    $student = Student::findOrFail($id);  
    return view('students.show', compact('student'));  
}
```

```
public function edit($id){  
    $student = Student::findOrFail($id);  
    return view('students.edit', compact('student'));  
}
```

```
public function update(Request $request, $id){  
    $student = Student::findOrFail($id);  
    $request->validate(['name' => 'required']);  
    $student->update($request->all());  
    return redirect()->route('students.index')->with('success', 'Student updated.');
```

```
    }  
    public function destroy($id){  
        $student = Student::findOrFail($id);  
        $student->delete();  
        return redirect()->route('students.index')->with('success', 'Student deleted.');
```



Web Route



```
// routes/web.php

use App\Http\Controllers\StudentController;

// Route::get('/students', [StudentController::class, 'index'])->name('students.index');
// Route::get('/students/create/', [StudentController::class, 'create'])->name('students.create');
// Route::post('/students', [StudentController::class, 'store'])->name('students.store');
// Route::get('/students/{id}', [StudentController::class, 'show'])->name('students.show');
// Route::get('/students/{id}/edit/', [StudentController::class, 'edit'])->name('students.edit');
// Route::put('/students/{id}', [StudentController::class, 'update'])->name('students.update');
// Route::delete('/students/{id}', [StudentController::class, 'destroy'])->name('students.destroy');

Route::resource('students', StudentController::class);
```

Blade Views: resources/views/students/ index.blade.php



```
<h1>Students</h1>
<a href="{{ route('students.create') }}">Add Student</a>
<ul>
@foreach ($students as $student)
    <li>
        {{ $student->name }} |
        <a href="{{ route('students.edit', $student->sid) }}">Edit</a> |
        <form method="POST" action="{{ route('students.destroy', $student->sid) }}" style="display:inline">
            @csrf @method('DELETE')
            <button type="submit">Delete</button>
        </form>
    </li>
@endforeach
</ul>
```



create.blade.php



```
<h2>Add Student</h2>
<form method="POST" action="{{ route('students.store') }}">
  @csrf
  <input name="name" placeholder="Name" required>
  <button type="submit">Save</button>
</form>
```



127.0.0.1:8000/students/create

Add Student

127.0.0.1:8000/students

Students

[Add Student](#)

- Sok | [Edit](#) |

edit.blade.php



```
<h2>Edit Student</h2>
<form method="POST" action="{{ route('students.update', $student->sid) }}">
    @csrf @method('PUT')
    <input name="name" value="{{ $student->name }}" required>
    <button type="submit">Update</button>
</form>
```



127.0.0.1:8000/students/1/edit

Edit Student



127.0.0.1:8000/students

Students

[Add Student](#)

- Sok Visal | [Edit](#) |

Web Authentication

Laravel Breeze

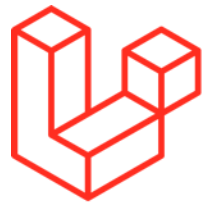
Install Laravel Breeze

```
composer require laravel/breeze --dev
```

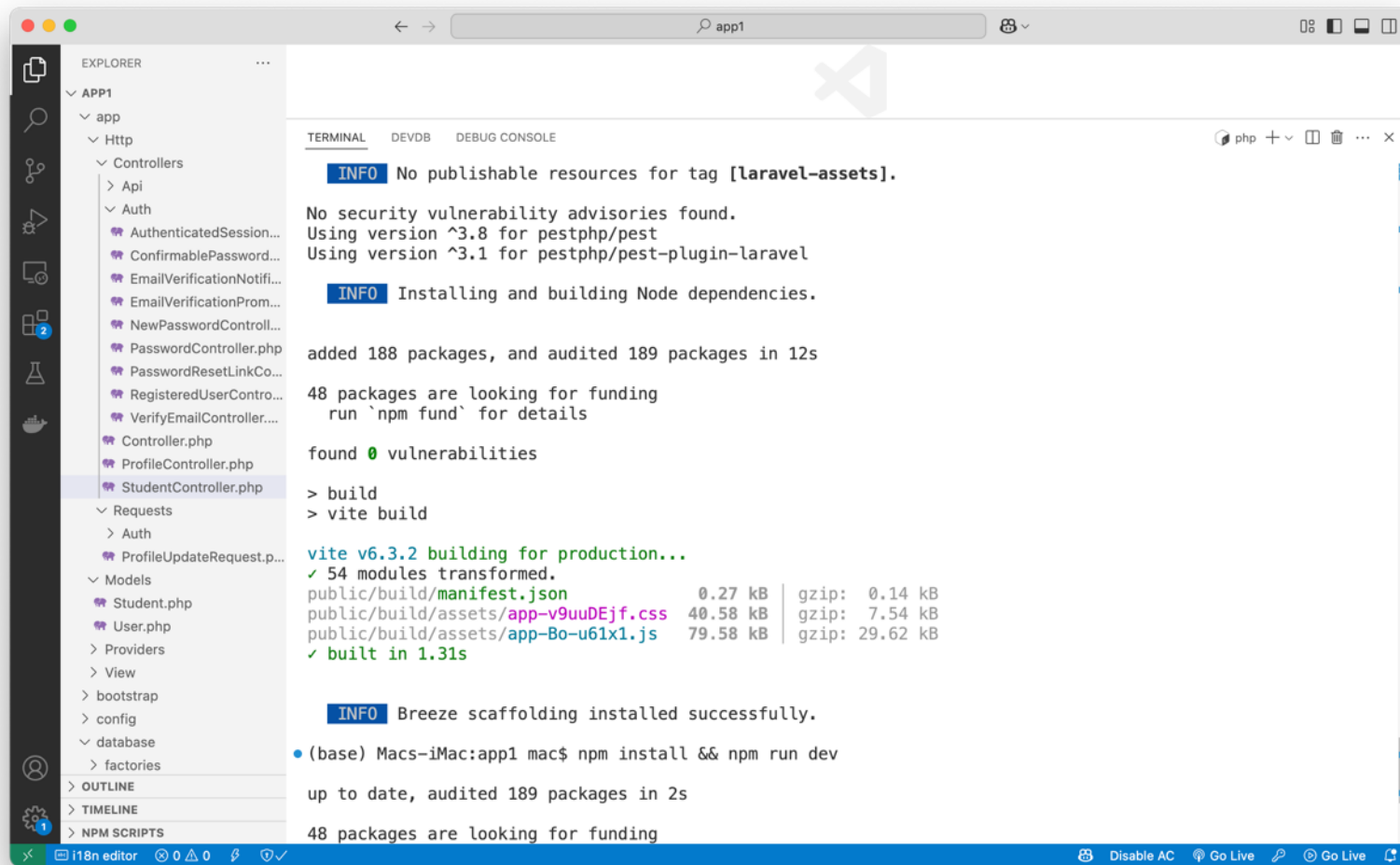
```
php artisan breeze:install
```

```
npm install && npm run dev
```

```
php artisan migrate
```



Laravel Breeze



Protect the student routes with auth middleware

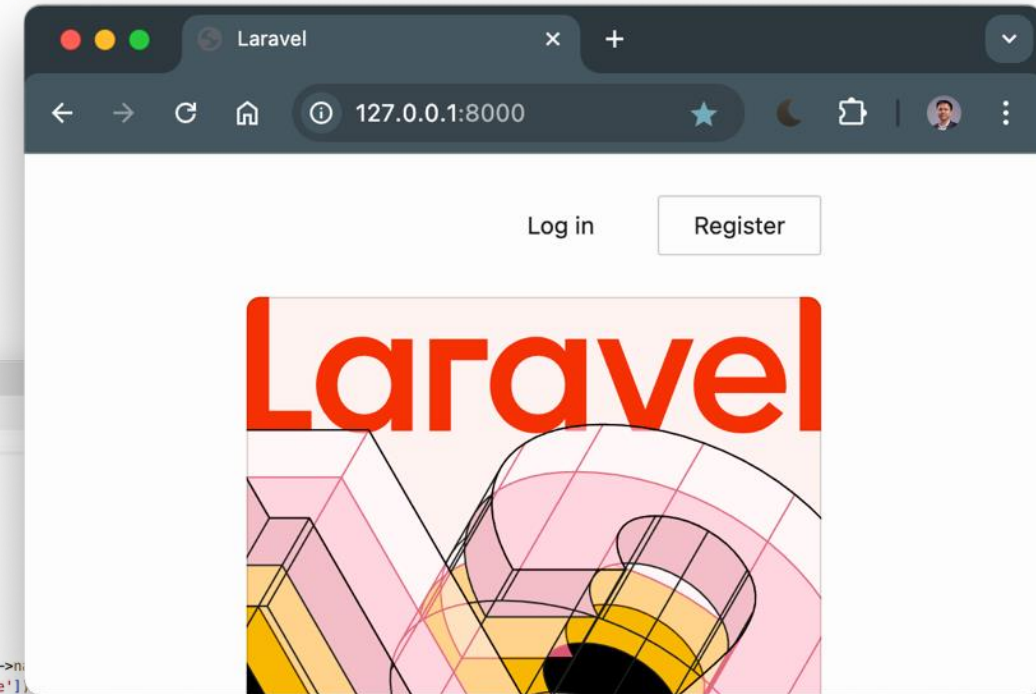
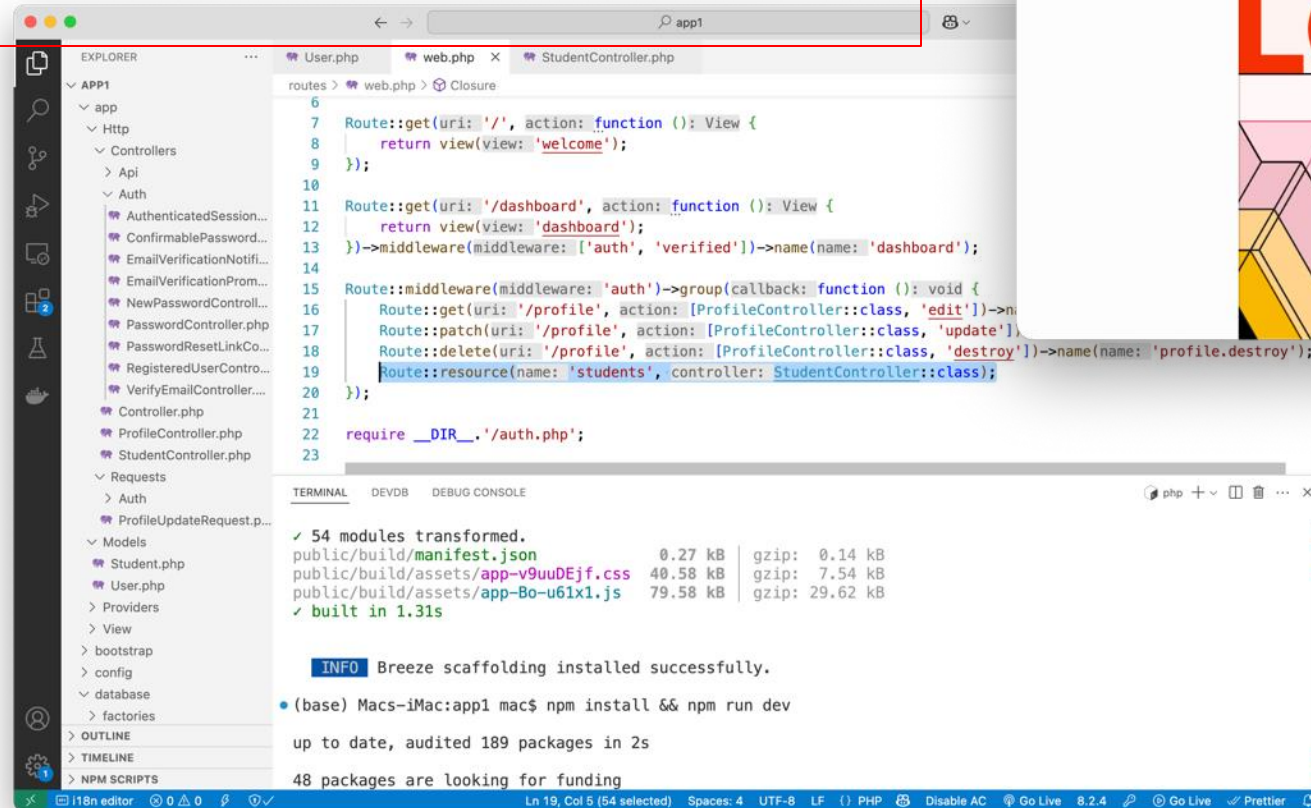


- In `routes/web.php`, wrap your student routes like this:

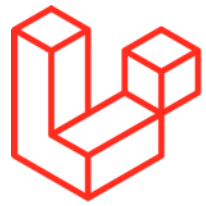
```
use App\Http\Controllers\StudentController;
```

```
Route::middleware(['auth'])->group(function () {  
    //other routes
```

```
    Route::resource('students', StudentController::class);  
});
```



When Accessing `/students`, it redirects to `/login`

A screenshot of a web browser window. The browser's address bar shows the URL '127.0.0.1:8000/login'. The page features the Laravel logo at the top center. Below the logo is a white login form with the following elements: an 'Email' label above a text input field, a 'Password' label above a text input field, a 'Remember me' checkbox, a '[Forgot your password?](#)' link, and a dark blue 'LOG IN' button.

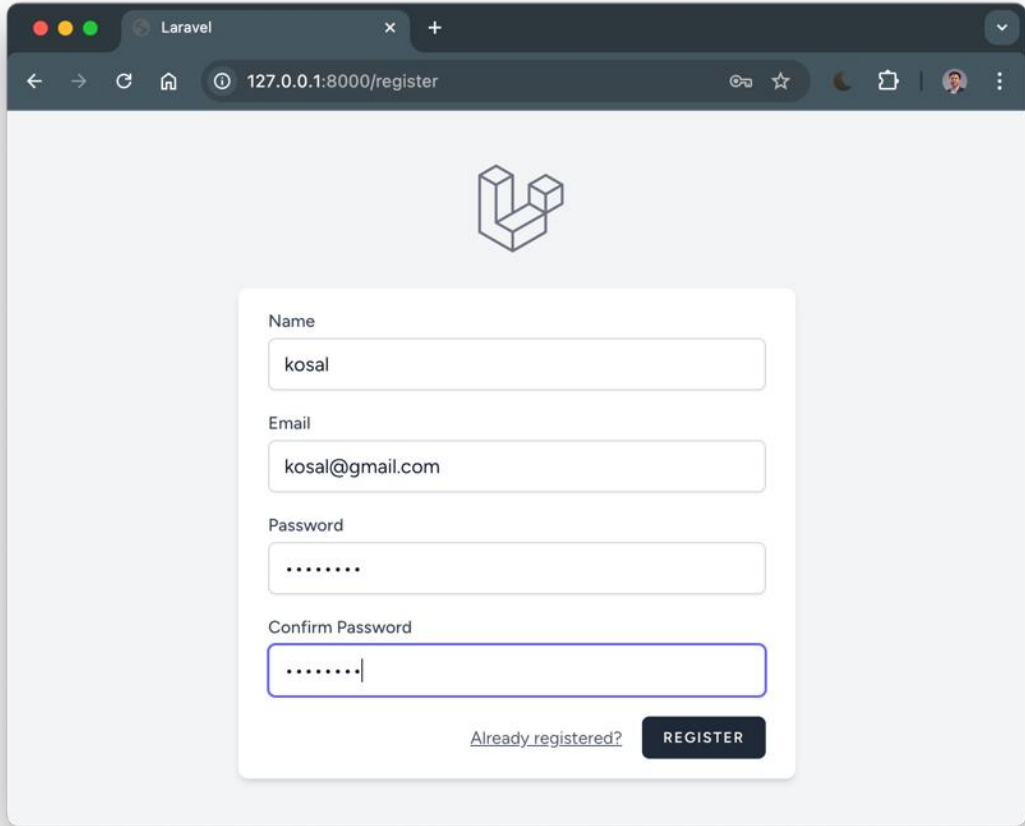
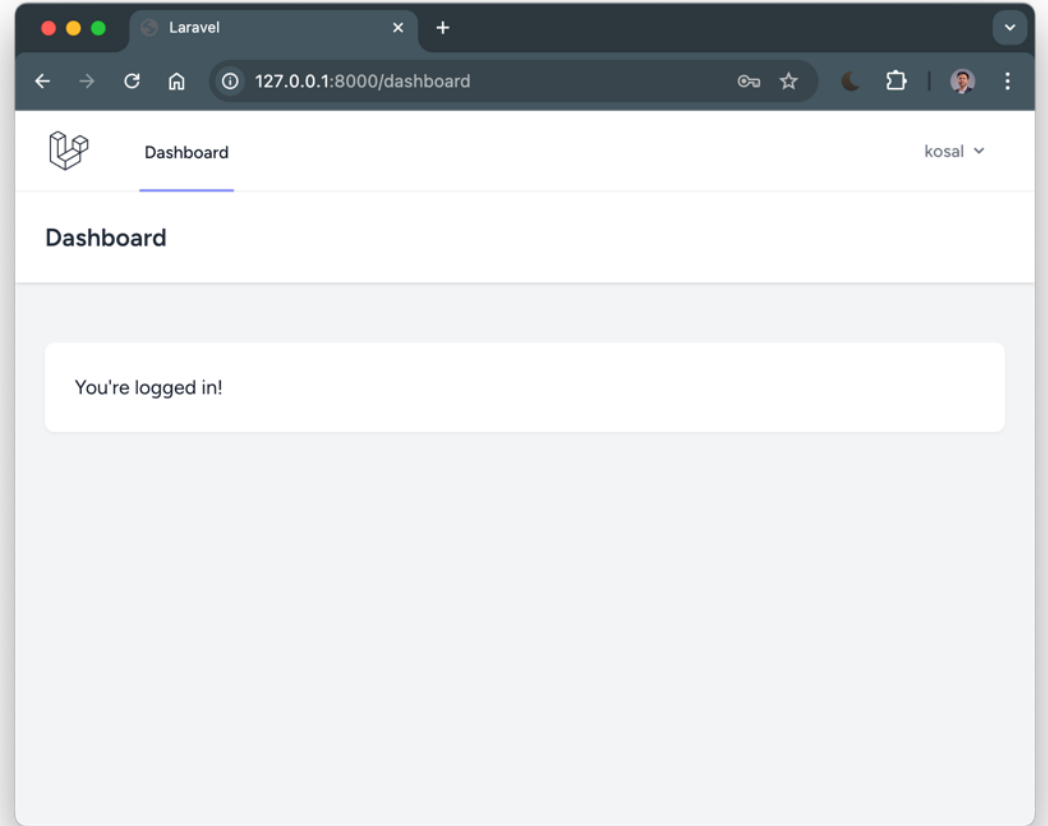
Email

Password

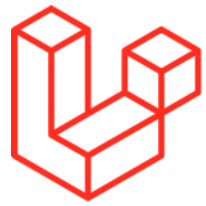
☐ Remember me

[Forgot your password?](#) **LOG IN**

Testing Register at /register

A screenshot of a web browser showing the Laravel registration page. The browser's address bar displays '127.0.0.1:8000/register'. The page features the Laravel logo at the top center. Below it is a registration form with four input fields: 'Name' (containing 'kosal'), 'Email' (containing 'kosal@gmail.com'), 'Password' (masked with dots), and 'Confirm Password' (also masked with dots). At the bottom of the form, there is a link that says 'Already registered?' and a dark blue button labeled 'REGISTER'.

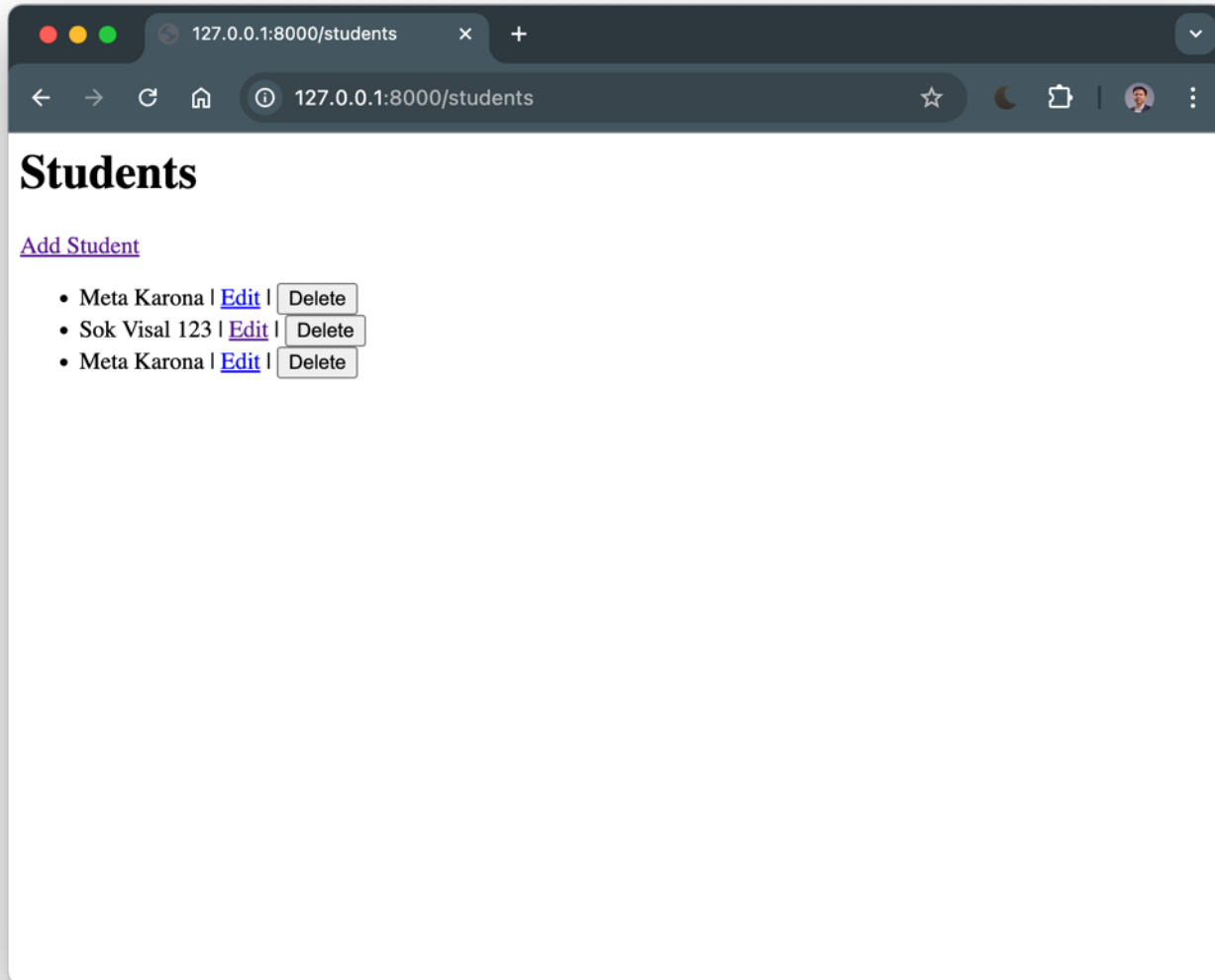
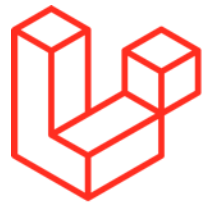
Editing Profile



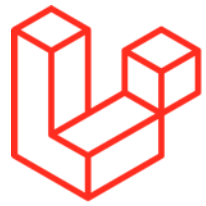
The screenshot shows a web browser window with the address bar displaying '127.0.0.1:8000/profile'. The page has a dark header with a 'Dashboard' link and a user profile 'kosal' with a dropdown menu containing 'Profile' and 'Log Out'. The main content area is titled 'Profile' and contains two sections: 'Profile Information' and 'Update Password'. The 'Profile Information' section has input fields for 'Name' (containing 'kosal') and 'Email' (containing 'kosal@gmail.com'), followed by a 'SAVE' button. The 'Update Password' section has input fields for 'Current Password', 'New Password', and 'Confirm Password'.

127.0.0.1:8000/profile

Now you can access /students



Add logout code in index.blade.php



```
<h1>Students</h1>
<!-- logout -->
<form method="POST" action="{{ route('logout') }}">
  @csrf
  <button type="submit">Logout</button>
</form>
<!-- end logout -->
<a href="{{ route('students.create') }}">Add Student</a>
<ul>
  @foreach ($students as $s)
    <li>
      {{ $s->name }} |
      <a href="{{ route('students.edit', $s->sid) }}">Edit</a> |
      <form method="POST" action="{{ route('students.destroy', $s->sid) }}" style="display:inline">
        @csrf @method('DELETE')
        <button type="submit">Delete</button>
      </form>
    </li>
  @endforeach
</ul>
```

